WEEK 2

Exploratory Data Analysis & Strategic Insights

Week 2: Exploratory Data Analysis & Strategic Insights
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Final Dataset: production_ready_dataset_v3_validated.xlsx

Total Records: 8,558

1. Introduction

The objective of Week 2 is to move beyond cleaning and feature engineering and perform a structured Exploratory Data Analysis (EDA) on Excelerate's learner–opportunity dataset. Building on the production-ready dataset from Week 1, we focus on:

- Understanding signup and engagement behaviour across time.
- Quantifying engagement patterns using features such as age, engagement lag, and opportunity duration.
- Visualizing trends, distributions, and outliers to support data-driven recommendations for later modeling.

The validated dataset contains 8,558 learner–opportunity records covering global learners (India, United States, Nigeria, Ghana, Pakistan, etc.), multiple opportunity categories (courses, workshops, competitions), and status outcomes such as Rejected, Team Allocated, Started, Dropped Out, and Applied.

1.1 Dataset Summary and Data Quality Revalidation

Overall Structure

- **Total records:** 8,558
- **Total columns:** 28
- Key analytical fields used in Week 2:
    - **Demographic:** age_years, gender, country, institution_name
    - **Temporal:** signup_month, signup_year, entry_created_at
    - **Engagement timing:** engagement_lag_days_fixed, days_before_start, engagement_lag_bucket
    - **Opportunity:** opportunity_duration_days, opportunity_category, status_description

2. Methodology

The analysis was performed using:

- Tools: Python (pandas, NumPy, matplotlib)
- Input dataset: production_ready_dataset_v3_validated.xlsx (Week 1 production-ready dataset with Week 2 validation)
- Core steps:
    - Re-validation of missing values and logical NaNs
    - Outlier and anomaly analysis for key numeric variables
    - Descriptive statistics (mean, median, percentiles, range)
    - Time-series inspection of signup activity
    - Engagement funnel analysis using status codes
    - Correlation analysis for age, lag, and duration metrics
    - Creation of multiple visualizations:
        - Boxplots, histograms, bar charts, line chart, and scatter plot

3. Data Quality Re-Validation

3.1 Missing Values and Logical NaNs

A fresh missing-value scan confirmed that the dataset remains highly complete after Week 1 cleaning:

- Top missing columns:
    - engagement_lag_days_fixed: 735 missing (8.59%)
    - age_years: 296 missing (3.46%)
    - signup_month: 295 missing (3.45%)
    - signup_year: 295 missing (3.45%)

All core identifiers and categorical descriptors such as opportunity_id, opportunity_name, opportunity_category, status_description, and country have 0% missing values.

For key numeric metrics:

- engagement_lag_days: 0% missing
- engagement_lag_days_fixed: 8.59% missing
  → These NaNs correspond to rows where the timeline was inverted in the raw data (apply_date < signup_date). Rather than forcing a value, these are intentionally set to NaN to avoid corrupting lag distribution.
- opportunity_duration_days: 0% missing
- days_before_start: 0% missing

These are logical NaNs, not data loss. They reflect situations where a metric cannot be reliably computed and must be documented, not “fixed” by arbitrary imputation.

3.2 “Unknown” Placeholders

Some categorical fields are populated with "Unknown" where no reliable source data exists:

- institution_name: 5 rows (0.06%)
- current_intended_major: 5 rows (0.06%)
- gender, country, status_description: 0 rows with "Unknown"

These are acceptable and clearly flagged. They should be treated as a separate category in analysis, not silently dropped.

4. Learner Profile: Age and Geography

4.1 Age Distribution

The engineered field age_years describes the age of each learner at signup:

- Count: 8,262 valid ages
- Mean: ~23.5 years
- Median: 23 years
- Min: 12 years
- 95th percentile: 32 years

- Max: 57 years

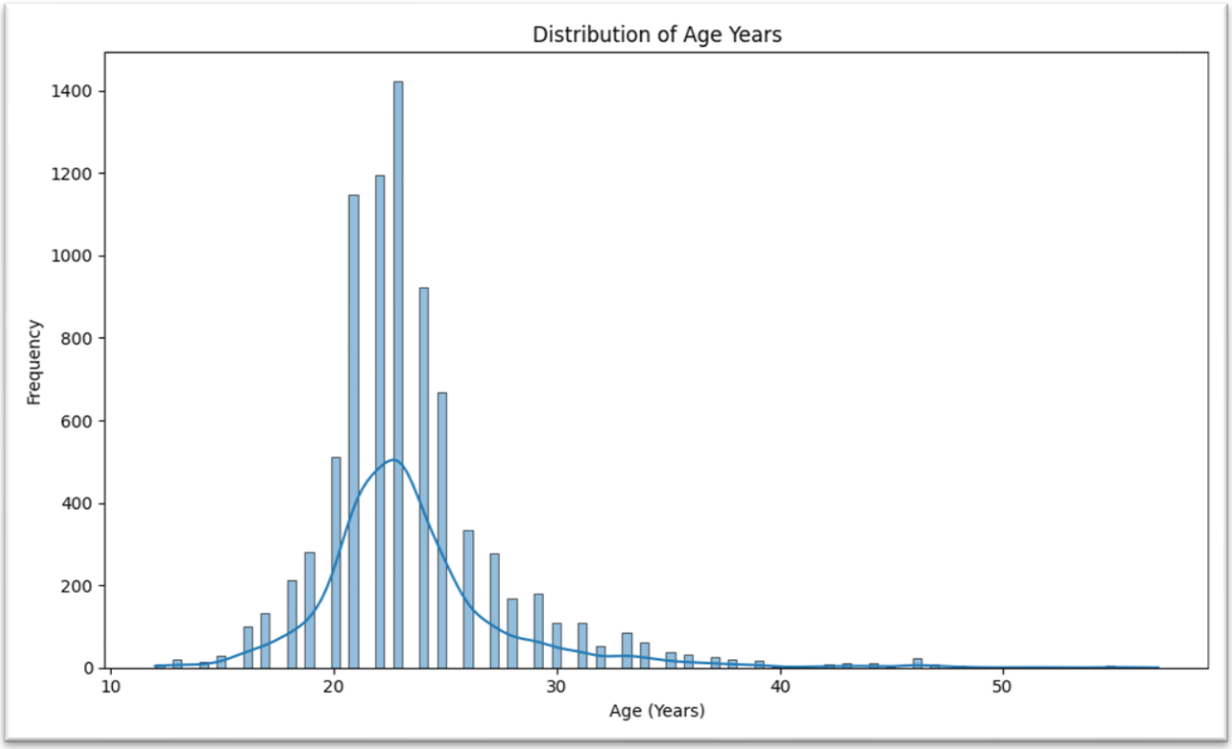


Figure 1: Age Distribution

The histogram (hist_age_distribution.png) shows a strong concentration between 21–25 years, confirming that the platform primarily serves university-age learners and early professionals. A small number of learners are younger (around 12–18) and older (35+); these will be important segments to monitor but do not represent data errors.

4.2 Country Distribution

The top countries by learner count are:

- United States: 3,976
- India: 2,836
- Nigeria: 760
- Ghana: 275
- Pakistan: 219

This confirms that the dataset is heavily skewed toward US and Indian learners, with a meaningful presence from Nigeria and other African countries. Any

modeling or recommendations must consider this geographic imbalance to avoid over-fitting to dominant regions.

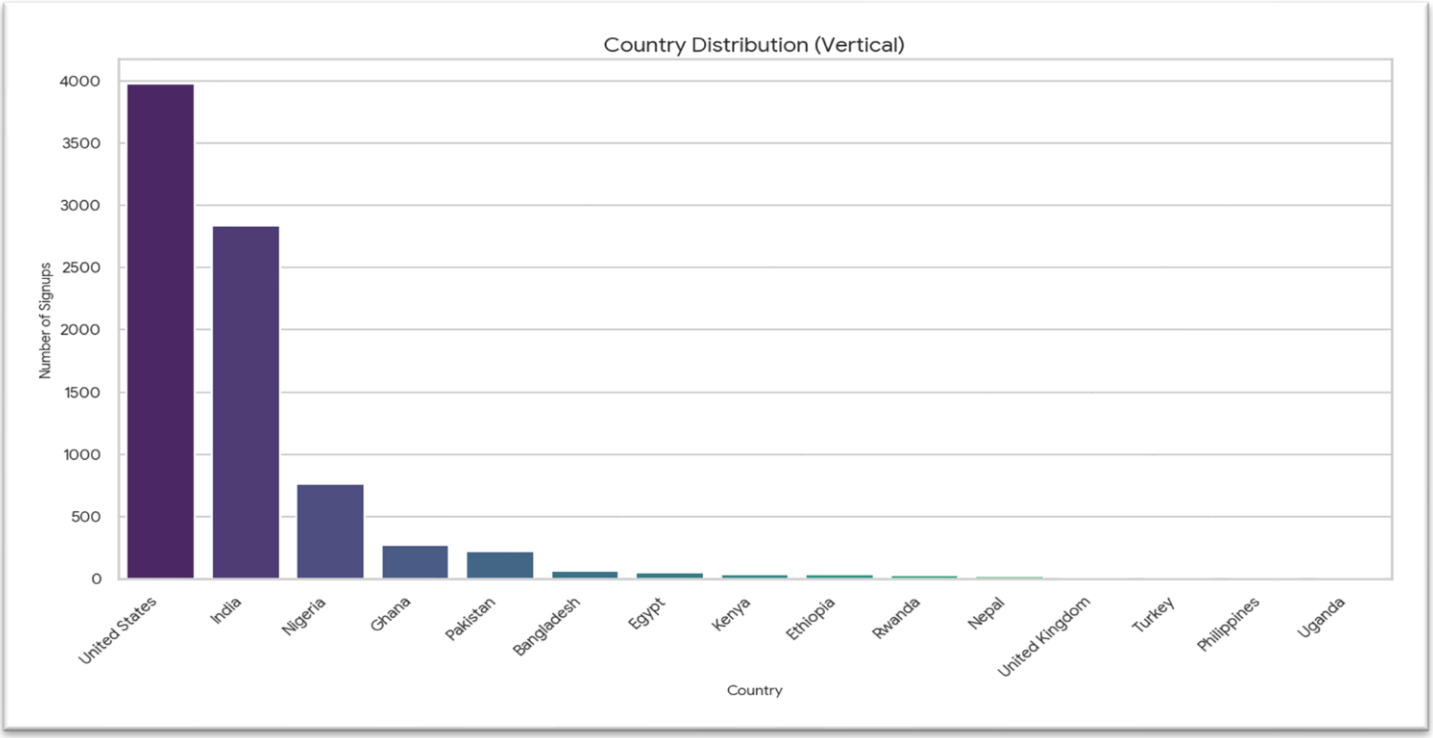


Figure 2: Country Distribution

5. Signup Trends Over Time

Signup behavior was analyzed using signup_year and signup_month, aggregated into a year_month field.

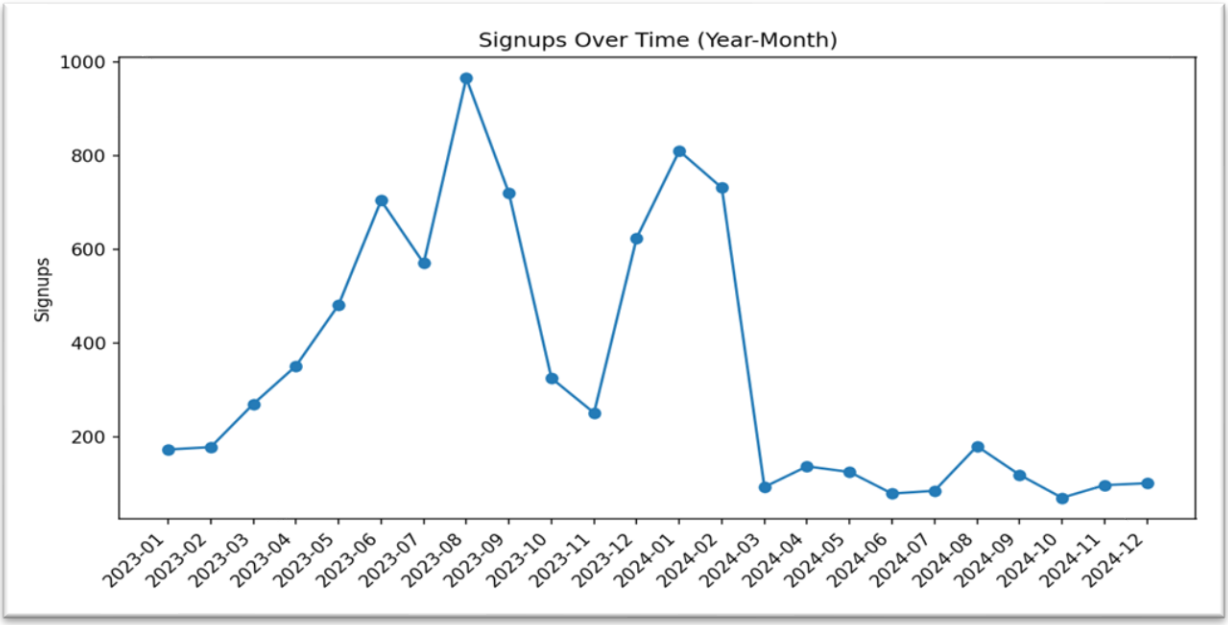


Figure 3: Signup Trends Over Time

Key observations from the time-series (line_signups_year_month.png):

- Strong signup activity throughout 2023, with clear peaks:
  - August 2023: 966 signups (largest single month)
  - High volumes from May–October 2023
- 2024 shows:
  - A strong start in January (811) and February (732)
  - Lower but still consistent activity in later months (e.g., August 2024: 181)
- Some records (~295) have no signup month/year (reflecting earlier missingness in source data), but they are a small minority.

Implication:
Excelerate's engagement is clearly seasonal, with strong peaks around mid-year and early in the academic cycle. These periods are ideal for targeted marketing campaigns and new program launches.

6. Engagement and Completion Status

6.1 Funnel Overview

The status_description field describes where each learner ends up in the opportunity lifecycle:
Status distribution:

- Rejected: 3,569 (41.70%)
- Team Allocated: 3,276 (38.28%)
- Started: 767 (8.96%)
- Dropped Out: 617 (7.21%)
- Waitlisted: 109 (1.27%)
- Applied: 105 (1.23%)
- Withdraw: 86 (1.00%)
- Rewards Award: 29 (0.34%)

This bar chart (bar_status_description.png) makes this pattern very clear visually.

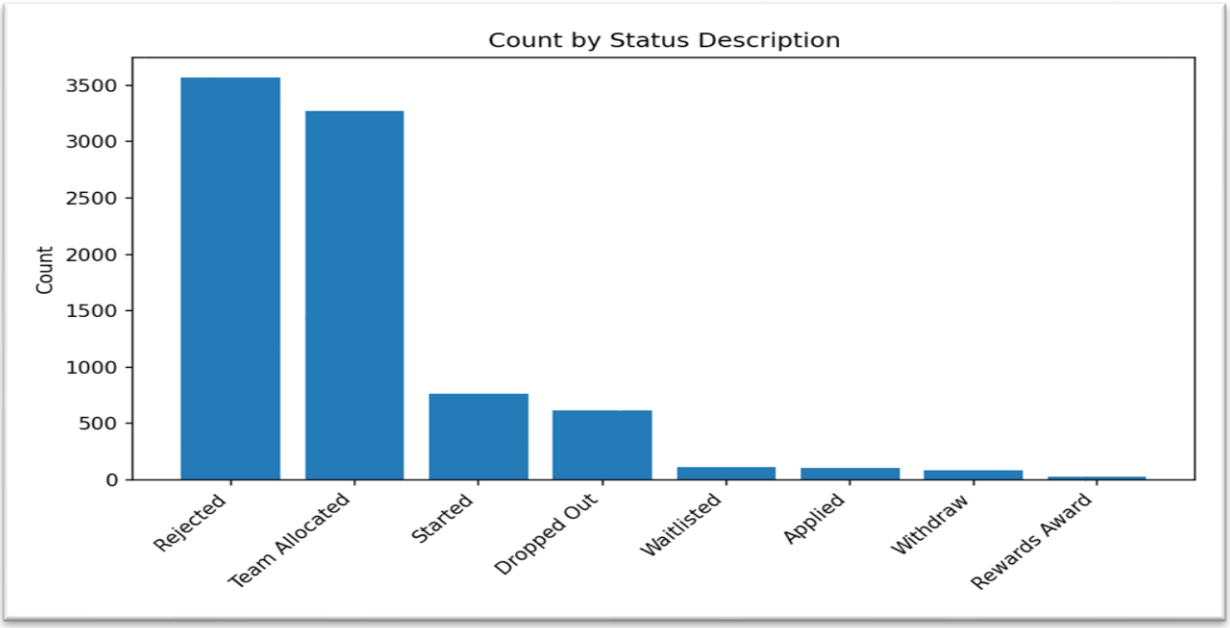


Figure 4: bar_status_description

For EDA, an engagement flag was defined:

- Engaged statuses: Started, Team Allocated, Rewards Award
- Engagement rate = proportion of rows with engaged_flag = 1

Result:

- Overall engagement rate: 47.58%

So, almost half of learners who appear in the dataset progress beyond simple application into an engaged state (team allocated, started, or recognition stage). However, the rejection rate is also very high (over 40%), and dropouts account for another ~7%.

6.2 Engagement Lag by Status

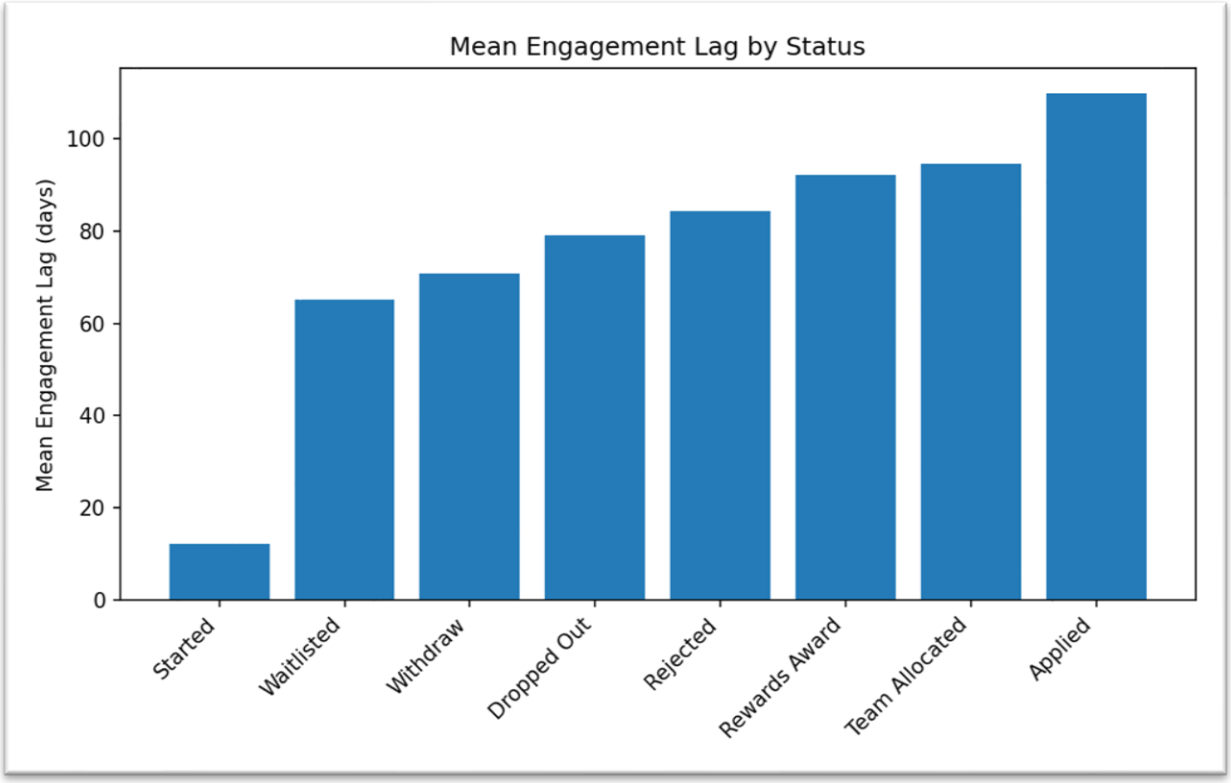


Figure 5: Engagement Lag by Status

Using engagement_lag_days_fixed, we computed mean lag by status and visualized it (bar_mean_lag_by_status.png). This helps answer:

"Do learners who get accepted or start earlier tend to apply sooner or later than others?"

This gives a starting point for Week 3–4 modeling around conversion and retention.

7. Behavioral Metrics: Engagement Lag, Days Before Start, Duration

7.1 Engagement Lag (engagement_lag_days_fixed)

This metric measures how many days pass between learner signup and application:

- Valid count: 7,823 (after setting inversion rows to NaN)
- Mean: 81.85 days
- Median: 1.0 day

- Min: 0 days
- 75th percentile: 142 days
- 95th percentile: 359 days
- Max: 695 days

Key points:

- The median of 1 day shows that most learners apply almost immediately after signup.
- The large mean (81.85 days) and high standard deviation (~127.6) indicate a long tail of learners who delay application for months.
- 376 learners have an engagement lag >365 days, meaning they waited more than a year to apply after signup.

Visuals:

- hist_engagement_lag_0_365.png (clipped 0–365 days) shows the skew toward low lag with a long tail.

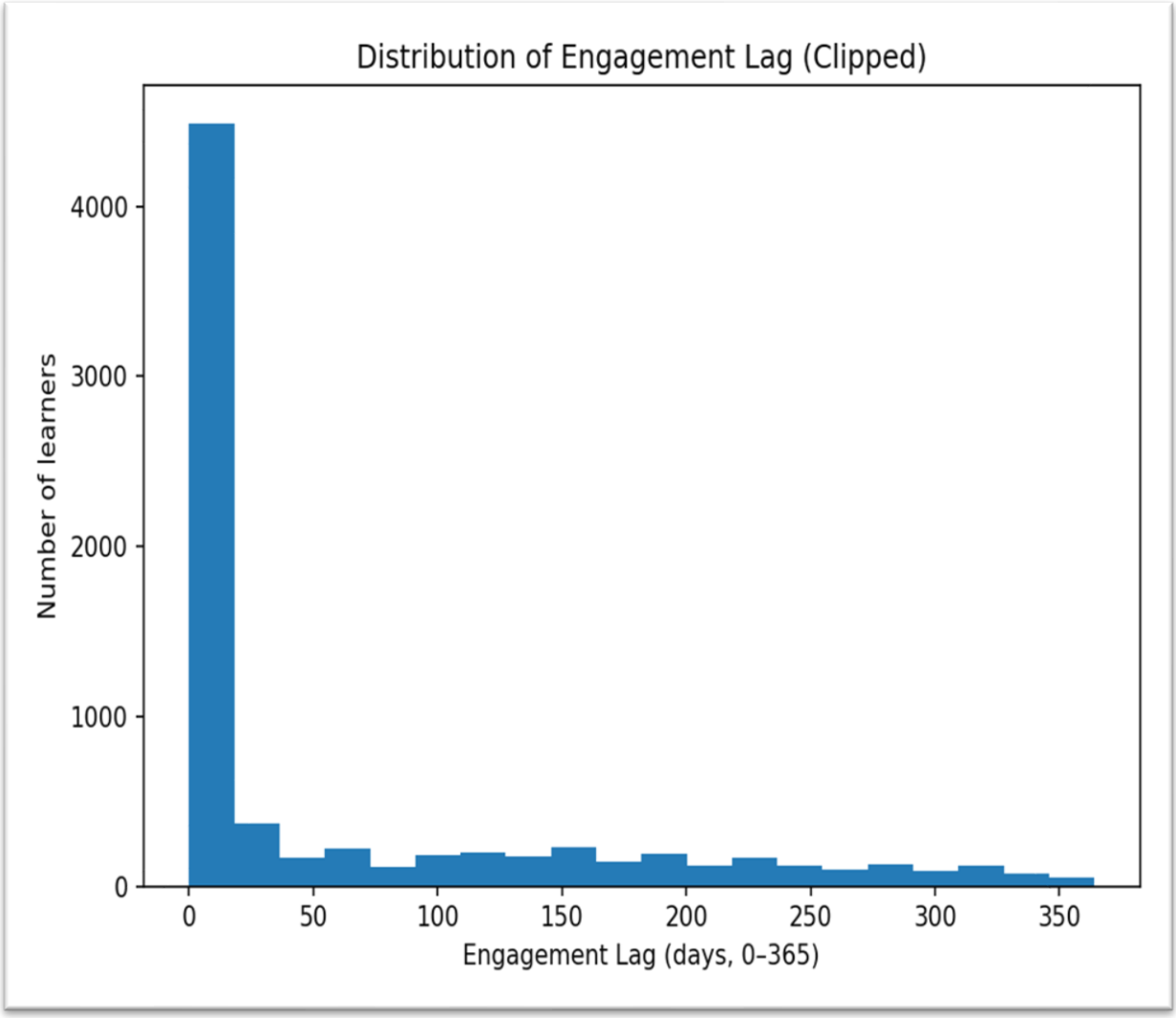


Figure 6: histogram of engagement lag

- Age vs. Engagement Lag (age_years vs. engagement_lag_days_fixed Scatter Plot):

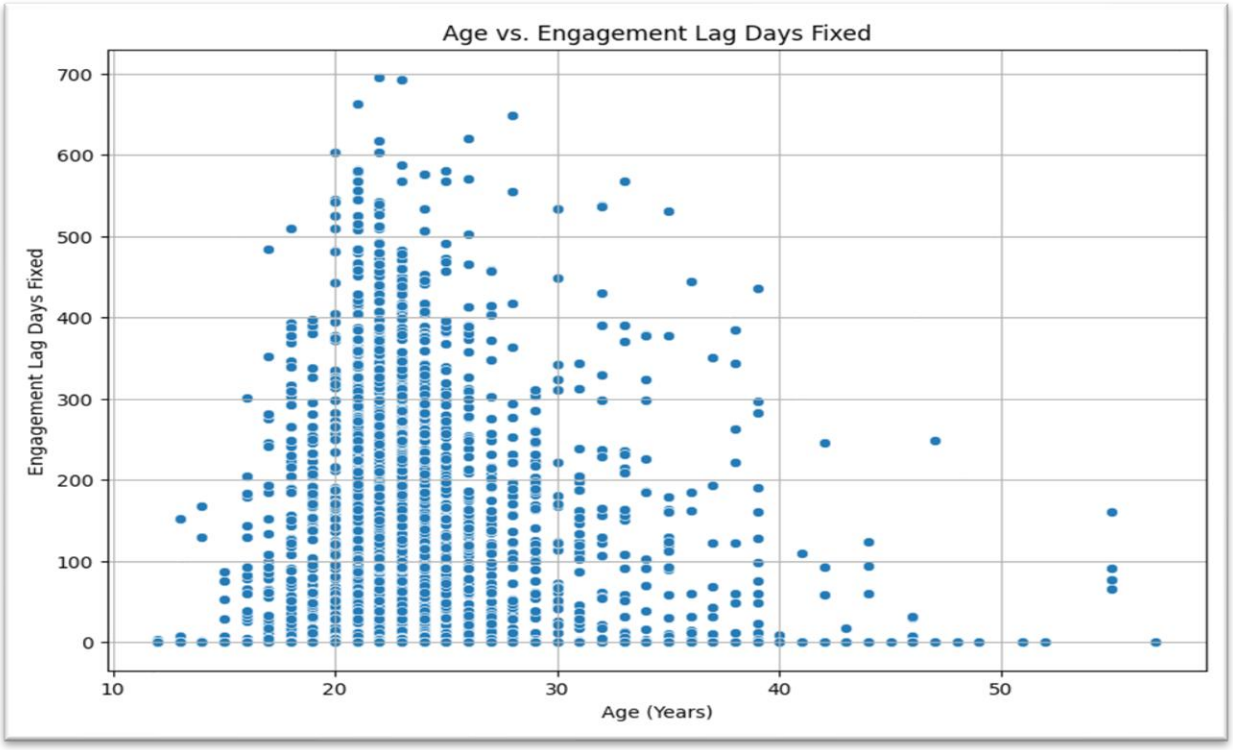


Figure 7: Age vs. Engagement Lag

- bar_engagement_lag_bucket.png groups learners into buckets (0, 1–7, 8–30, 31–90, 90+, Unknown), making it easier to talk about "fast", "medium", and "very slow" engagers.

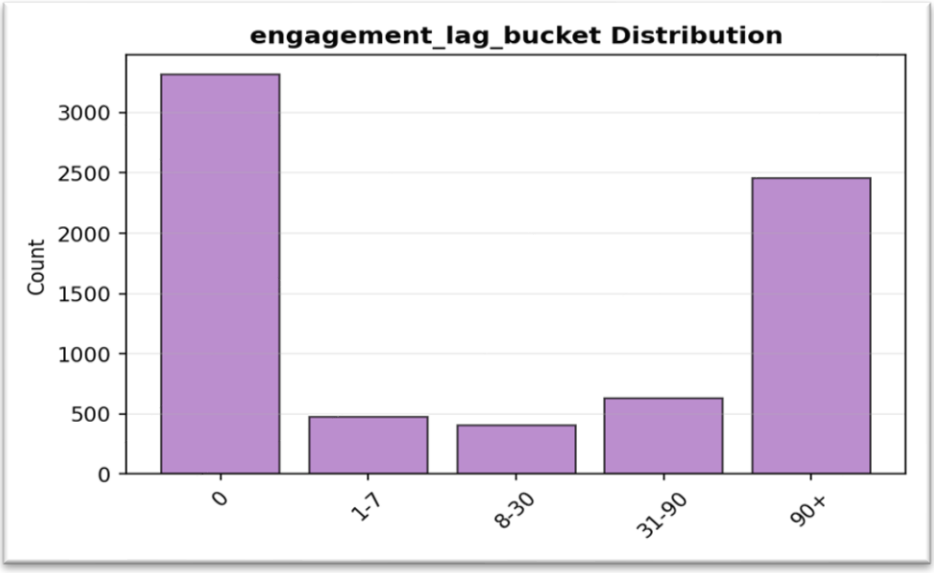
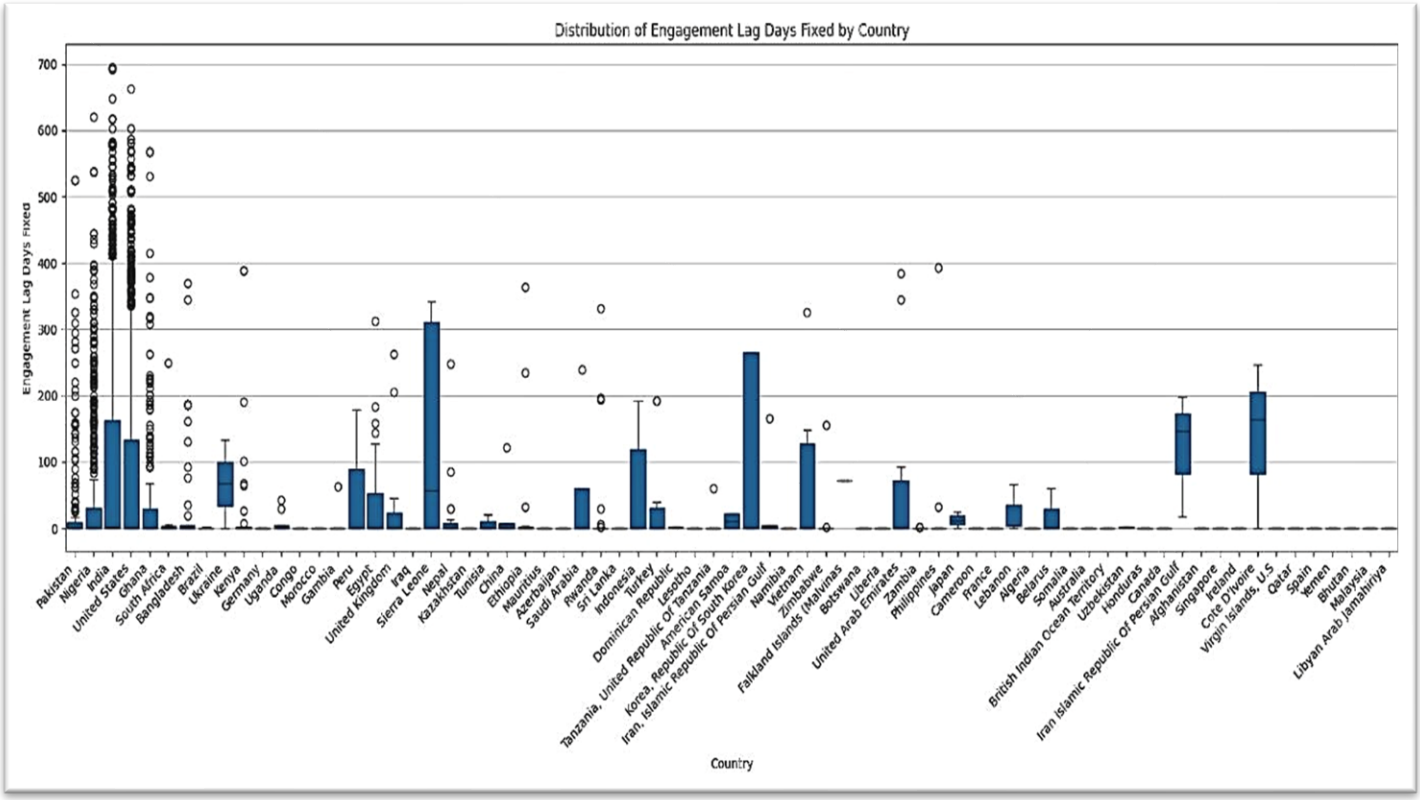


Figure 8: bar graph of engagement lag bucket

- **Country vs. Engagement Lag (engagement_lag_days_fixed by country BoxPlot):**



- **Figure 9: engagement lag days by country Boxplot**

Implication:
There are distinct engagement segments:

- Immediate engagers (within days).
- Slow burn users (months).
- Very late re-activations (over a year).

These should be treated as separate behavioral cohorts in future analysis and communication strategies.

7.2 Days Before Start (days_before_start)

This feature captures the difference between opportunity start date and application date:

- Mean: 30.28 days

- Median: 0 days
- Max: 445 days

A median of 0 indicates that many learners apply very close to the opportunity start date (on or just before). A subset applies far in advance, and flagged extremes (flag_days_before_start_extreme) highlight learners more than a year away from the start date in either direction. These cases are valid but unusual, and should be treated as distinct behavioral patterns rather than errors.

7.3 Opportunity Duration (opportunity_duration_days)

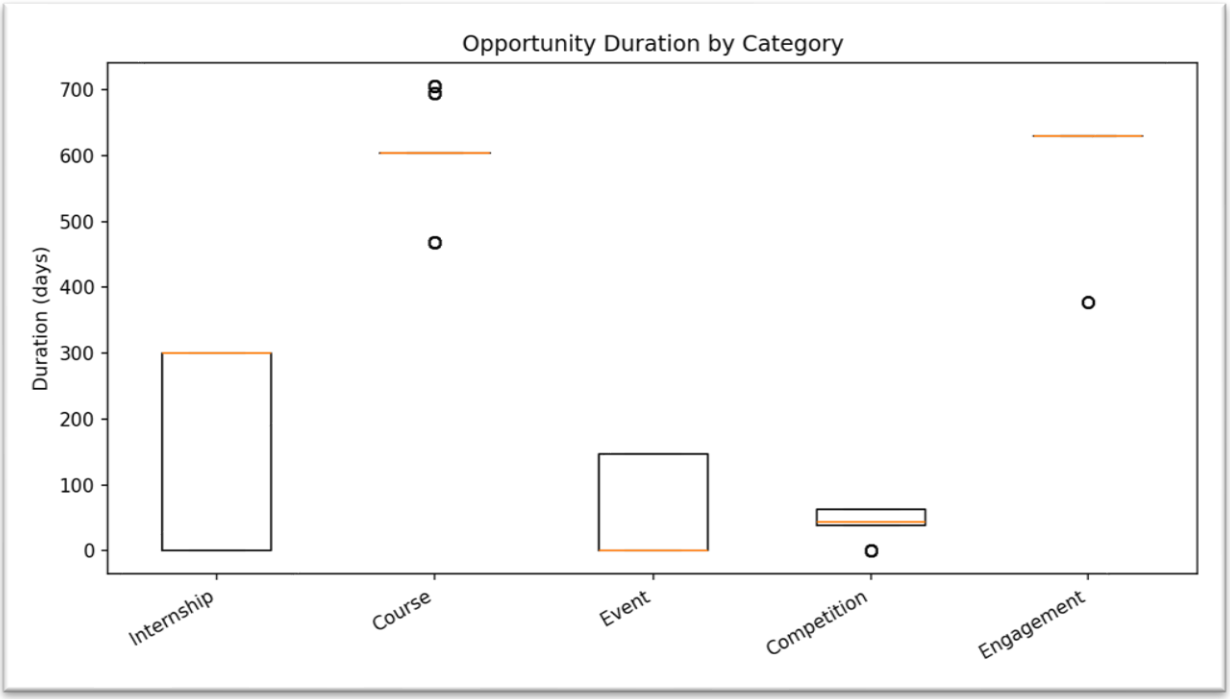
This represents the length of each opportunity:

- Range: 0 to 706 days
- Mean: ~253 days
- Median: 300 days
- 95th percentile: 604 days
- Max: 706 days
- Zero-duration entries: 2,994 rows (duration_days == 0)

Highlights:

- 33 records have durations >700 days, all at 706 days, concentrated in a specific long-running program (Jump Start: Developing Your Emotional Intelligence).
- Zero-duration rows typically correspond to opportunities with unknown end dates or special record logic (e.g., open-ended or missing end date). They are correctly left as 0 and must be interpreted carefully.

The boxplot by category (boxplot_duration_by_category.png) compares duration across opportunity_category (e.g., Courses, Events, Competitions) and shows



which formats tend to be short vs long.

Figure 10: boxplot duration by category

8. Patterns, Correlations, and Relationships

A correlation matrix was computed for the main numeric features (age_years, engagement_lag_days_fixed, opportunity_duration_days, days_before_start):

- Relationships are generally weak in magnitude:
  - engagement_lag_days_fixed vs days_before_start: about -0.19 (moderate negative)
    → Learners who apply a long time before start tend to show shorter lag between signup and application, which is intuitive.
  - age_years has minimal correlation with other variables (close to 0), suggesting age does not strongly drive engagement lag or duration at this level of aggregation.
  - opportunity_duration_days shows small negative correlations with age and days_before_start, but not strong enough to justify major claims yet.

The scatter plot (scatter_age_vs_engagement_lag.png) visually confirms:

- No obvious strong trend between age and engagement lag: younger and older learners both appear across the full lag spectrum.

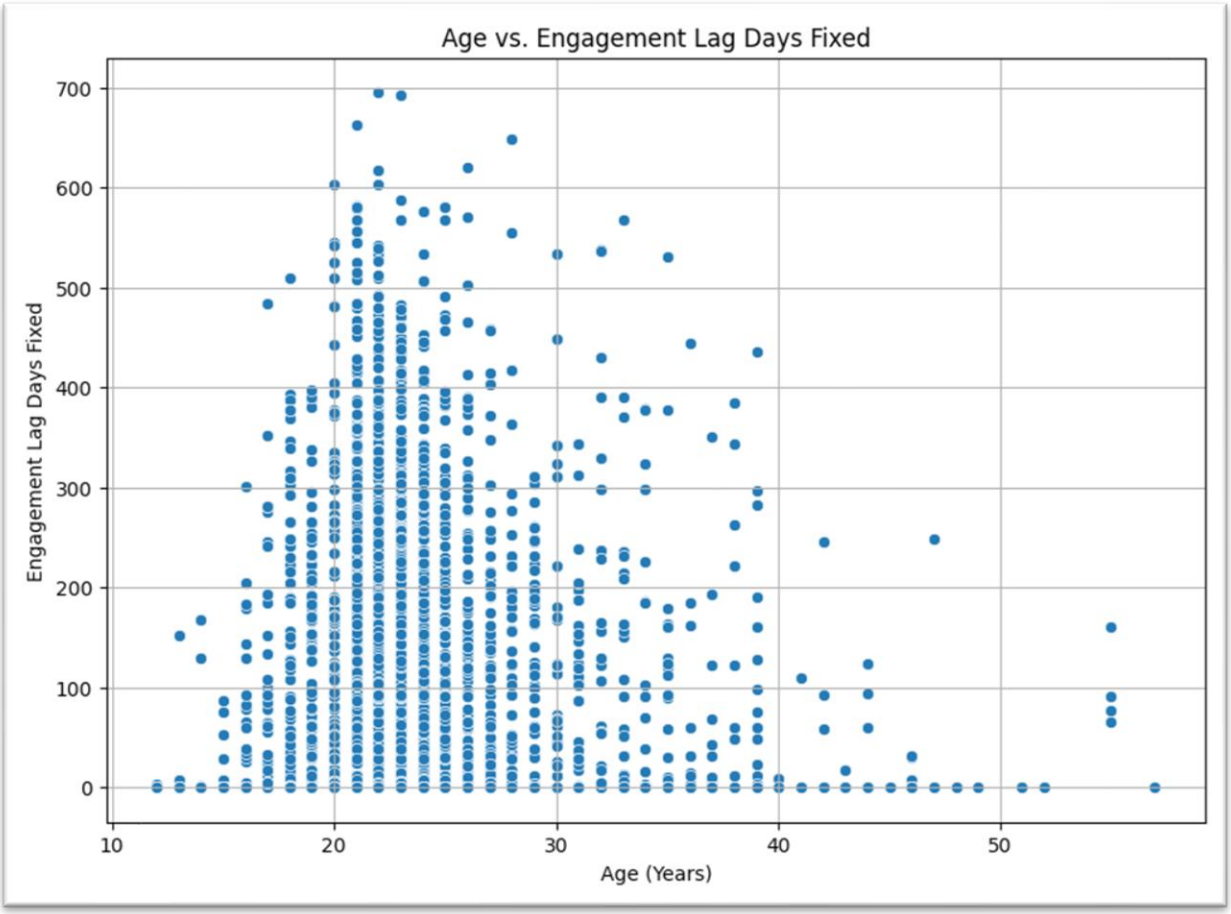


Figure 11: scatter Graph of age vs engagement_lag

9. Outliers and Anomalies

9.1 Engagement Inversions

- 735 records are flagged with flag_engagement_inversion = 1, indicating apply_date < signup_date in the raw data.

- For these flagged rows, engagement_lag_days_fixed is set to NaN, and:
    - Flagged rows with lag = NaN: 735 (expected)
    - Flagged rows with lag NOT NaN: 0 (correct)
    - Unflagged rows with negative lag: 0 (no hidden problems)

These records are preserved for integrity, but their lag metric is excluded from numerical analysis to avoid distortion.

9.2 Extreme Days Before Start

- 394 rows are flagged by flag_days_before_start_extreme, representing very early or very late applicants relative to the opportunity start date.
- These are not deleted; they represent important edge behavior (e.g., very proactive learners or very late applicants).

9.3 Age and Duration Extremes

- Age extremes (12 and 57) are rare but plausible and consistent with earlier validation (unrealistic ages >120 were already removed in Week 1).
- Duration extremes (706 days) correspond to a specific long program and should be treated as valid, long-form engagements, not errors.

Regional Rejection Hotspots: The Segmentation Heatmap identified that applicants from Nigeria and Pakistan face disproportionately higher rejection rates compared to US applicants. This suggests a potential misalignment in qualification criteria or application support for these regions.

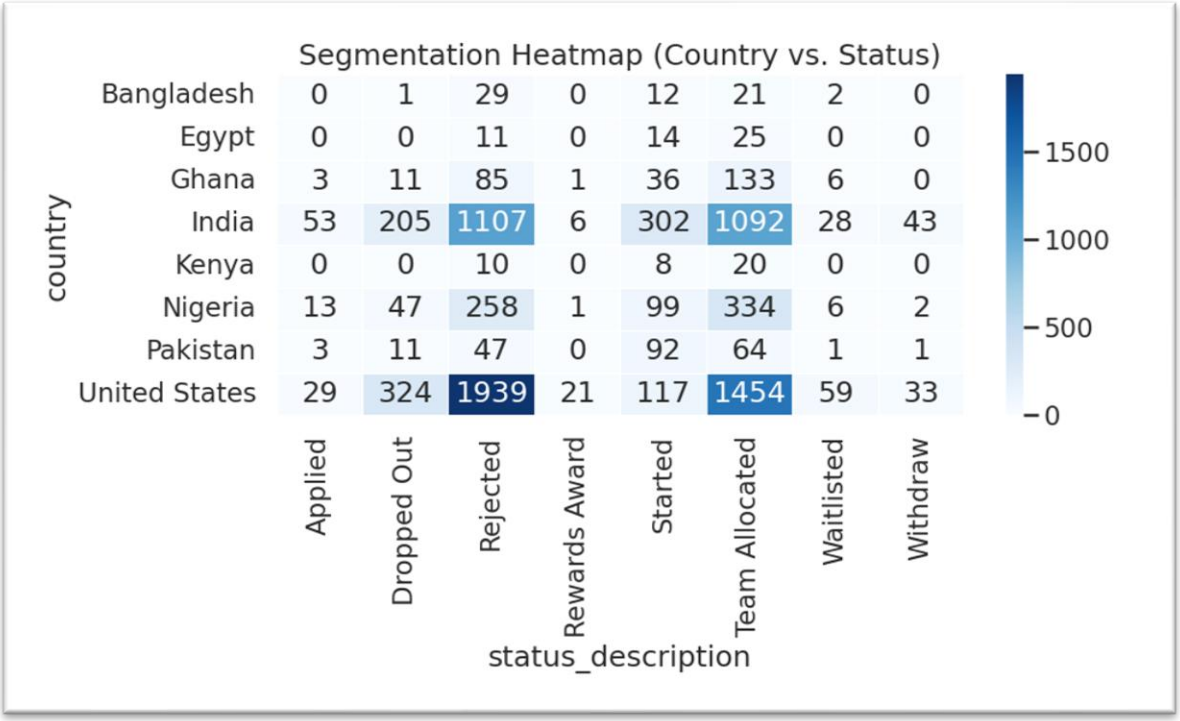


Figure 12: Segmentation Heatmap

10. Key Insights and Recommendations

10.1 Signup and Engagement

- Signup activity is seasonal. Peaks around August 2023 and early 2024 should be treated as strategic windows for campaigns and new offerings.
- Engagement rate is ~47.58%. Nearly half of learners end up in positive engagement states (Started / Team Allocated / Rewards Award), but the rejection rate is over 40%, which signals either strong filtering or mismatched expectations.

Recommendation:
Review criteria and communications around opportunities with very high rejection rates. Improving pre-screening or guidance could reduce wasted applications and increase satisfaction.

10.2 Lag Behavior

- The distribution of engagement lag is very skewed:

    - Many learners act immediately.
    - A significant tail waits months or even a year+.

Recommendations:

- Treat fast, medium, and slow engagers as separate segments when designing nudges, reminders, and follow-ups.
- For slow and very slow engagers, test re-activation campaigns, targeted reminders, or evergreen opportunities.

10.3 Program Design and Duration

- Long programs (600–700+ days) may require special retention strategies and staggered milestones to keep learners motivated.
- Zero-duration records (2994 rows) should be interpreted as "missing/unknown end date" or "open-ended", and handled carefully in downstream analyses.

Recommendation:
For modeling and BI dashboards, clearly label zero duration as "Unknown/Unspecified" instead of treating them as genuinely 0-day programs.

10.4 Data Strategy

- The choice to preserve anomalies (inversions, extreme timing) and to use flags + NaNs instead of silent fixes is correct.
- These flags should be:
    - Used in future models as features (e.g., flag_engagement_inversion, flag_days_before_start_extreme).
    - Exposed in BI dashboards to support data quality monitoring.

11. Conclusion and Next Steps

The Week 2 EDA confirms that:

- The dataset is structurally sound and analytically usable, with clearly documented logical NaNs and flags.
- We have a solid understanding of:
  - Who the learners are (age, geography).
  - When they sign up (time-based trends).
  - How they move through the funnel (status distribution and engagement rate).
  - How long they take to engage (engagement lag).
  - How long opportunities last (duration patterns and extremes).

By preparing this report and dataset, interns provide a comprehensive summary of their work, setting the stage for effective analysis and modeling in the following weeks. The next logical steps are:

- Deeper cohort analysis (by country, institution, opportunity category).
- Feature selection and transformation for predictive models on:
  - Likelihood of engagement (Started / Team Allocated / Rewards Award).
  - Risk of dropout or non-completion.
- Dashboard creation to operationalize these insights for business stakeholders.

This EDA completes the Week 2 objectives: the data is not only clean and validated, but also well-understood, with clear, visual evidence to support decision-making and future modeling work.